INCH-POUND

MIL-DTL-22200/1H(SH)

w/ INT. AMENDMENT 1
19 April 2010

USED IN LIEU OF
MIL-DTL-22200/1H(SH)
22 May 2008

DETAIL SPECIFICATION SHEET

ELECTRODES, WELDING, MINERAL COVERED, IRON-POWDER, LOW-HYDROGEN MEDIUM AND HIGH TENSILE STEEL, AS-WELDED OR STRESS-RELIEVED WELD APPLICATION

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-E-22200.

1. SCOPE

1.1 Scope. This specification covers iron-powder, low-hydrogen types of mineral covered electrodes, for the fabrication and repair welding of low and medium carbon and high tensile steels for as-welded and stress-relieved application.

1.2 Classification. Iron-powder, low-hydrogen electrodes are furnished in the types, classes, positions of welding, and sizes as specified in table I (see 6.2).

TABLE I. Types, classes, positions of welding, and sizes.

Type 1/ 2/ 3/	Class	Positions of welding	Sizes (diameter, inches)
MIL-7018-M MIL-8018-C3 MIL-8018-C3SR	1	All	3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 7/32, 1/4
	3	Flat	5/16
MIL-9018-M MIL-11018-M	1	All	3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 7/32, 1/4

NOTES:

1/ Type classifications of electrodes will be determined according to the capability of the electrode to meet the tests as specified herein, when used in the positions of welding as specified in table I and section 4 herein. An electrode will be approved under only one type classification, and an electrode will not be approved if it differs in type from that stated in the manufacturer's request for test. The position description indicates the maximum size qualified under the specification. However, reference should be made to the Qualified Products List (QPL) for the selection of electrodes for all positions of welding.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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TABLE I. Types, classes, positions of welding, and sizes - Continued.

NOTES – Continued:

2/ The suffix RC added to the MIL-type pertains to restriction of the copper content (see 3.5). All requirements, tests, and acceptance criteria remain otherwise unchanged.

3/ Addition of the suffix SR to the basic MIL-type designation indicates a special MIL-type electrode formulated to meet the long time (50 hours) stress-relieved mechanical properties/requirements only.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-E-22200	-	Electrodes, Welding, Covered: General Specification for
MIL-S-22698	-	Nondestructive Testing Acceptance Criteria

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035	-	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVSEA TECHNICAL PUBLICATIONS

T9074-AS-GIB-010/271	-	Requirements for Nondestructive Testing Methods
T9074-BD-GIB-010/0300	-	Base Materials for Critical Applications: Requirements for Low-Alloy Steel Plate, Forgings, Castings, Shapes, Bars, and Heads of HY-80/100/130 and HSLA-80/100

(Copies of these documents are available from the Naval Logistics Library, 5450 Carlisle Pike, Mechanicsburg, PA 17055 or online at http://nll.ahf.nmci.navy.mil.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NCSL Z540-1	-	Calibration Laboratories and Measuring and Test Equipment-General Requirements (DoD adopted)

(Copies of this document are available from the American National Standards Institute, 25 W. 43rd St, 4th Floor, New York, NY 10036 or online at http://webstore.ansi.org/)

AMERICAN WELDING SOCIETY (AWS)

AWS A4.3	-	Standard Methods for Determination of the Diffusible Hydrogen Content of Martensitic, Bainitic, and Ferritic Steel Weld Metal Produced by Arc Welding (DoD adopted)
AWS A4.4M	-	Standard Procedures for Determination of Moisture Content of Welding Fluxes and Welding Electrode Flux Coverings
AWS A5.5/A5.5M	-	Specification for Low-Alloy Steel Electrodes for Shielded Metal Arc Welding (DoD adopted)
AWS B4.0	-	Standard Methods for Mechanical Testing of Welds (DoD adopted)

(Copies of these documents are available from the American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33126 or online at www.aws.org.)

ASTM INTERNATIONAL

ASTM A36/A36M	-	Standard Specification for Carbon Structural Steel (DoD adopted)

(Copies of this document are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specification exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Electrodes furnished under the specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Electrodes. Electrodes furnished under this specification shall conform to the requirements of MIL-E-22200 and as specified herein.

3.3 Covering composition and iron content.

3.3.1 Composition. The chemical composition of the coverings, except as specified in 3.3.2, is optional with the manufacturer. The water content of the coverings shall not exceed the percent-by-weight values as specified in table II. In addition, for MIL-7018-M and MIL-11018-M electrodes, the diffusible hydrogen levels in milliliters per one hundred grams (mL/100g) of deposited weld metal shall not exceed the following values for welds deposited with electrodes removed from sealed manufacturer's containers (see 4.6.1):

a. Electrodes ⅛-inch size and smaller, maximum average value of 3.2 mL/100g and maximum single value of 4.0 mL/100g.

b. Electrodes $^{5}/_{32}$-inch size and larger, maximum average value of 3.5 mL/100g and maximum single value of 4.3 mL/100g.

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TABLE II. Water content.

Type	Maximum water content (percent by weight) [1/]	
	Condition A [2/]	Condition B [3/, 4/]
MIL-7018-M	0.10	0.20
MIL-8018-C3	0.10	0.20
MIL-8018-C3SR	0.10	0.20
MIL-9018-M	0.10	0.20
MIL-11018-M	0.10	0.20

NOTES:

1/ If the infrared detection method identified in AWS A4.4M is used for determining moisture content, Condition A tests shall display a maximum water content of 0.15 percent and Condition B tests shall display a maximum water content of 0.2 percent. For test methods other than infrared detection, the limits as specified in table II apply.

2/ Condition A - When removed from sealed manufacturer's container.

3/ Condition B - After 9 hours –0, +30 minutes of exposure at 80 percent relative humidity (RH) and 80 °F as specified in 4.8.

4/ For receipt inspection by the contracting activity, the maximum water content shall not be greater than 0.25 percent for all methods except the infrared method which shall not be greater than 0.29 percent.

3.3.2 Total iron content. The total iron content of the covering, including any combined iron in addition to metallic iron powder, shall be not less than 15 percent.

3.4 Groove welds.

3.4.1 Mechanical properties. The weld metal properties for the tensile and impact tests required for qualification inspection, conformance inspection or the acquisition document shall be as specified in table III. When specified (see 6.2), the welded test assembly shall be stress-relieved as stated in 4.4 and as follows:

a. MIL-7018-M, MIL-8018-C3, and MIL-8018-C3SR. For MIL-7018-M, the welded test assembly shall be heated to 1150±25 °F, held at this temperature for one hour and furnace cooled at 200 °F per hour or slower to below 500 °F. For MIL-8018-C3 and MIL-8018-C3SR, the welded test assembly shall be heated to 1150±25 °F, held at this temperature for two hours and furnace cooled at 200 °F per hour or slower to below 500 °F.

b. MIL-8018-C3SR. The welded test assembly shall be heated to 1150±25 °F, held at this temperature for 50 hours minimum and furnace cooled at 10 °F per hour or slower to below 600 °F.

c. MIL-9018-M and MIL-11018-M. The welded test assembly shall be heated to 1025±25 °F, held at this temperature for one hour and furnace cooled at 200 °F per hour or slower to below 500 °F.

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TABLE III. Mechanical properties. [1/]

Tensile						
Type	Ultimate tensile strength		Yield strength [2/]		Elongation [3/]	
	As-welded	Stress-relieved	As-welded [4/]	Stress-relieved	As-welded	Stress-relieved
	Minimum psi	Minimum psi	Range, psi	Minimum psi	Minimum (percent)	Minimum (percent)
MIL-7018-M	[5/]	----	53,000 to 75,000	51,000	24	24
MIL-8018-C3	80,000	----	68,000 to 80,000	65,000	24	24
MIL-8018-C3SR	----	80,000	----	65,000	----	24
MIL-9018-M	90,000	----	78,000 to 90,000	75,000	24	24
MIL-11018-M	[5/]	----	98,000 to 110,000	95,000	20	20

Charpy V-notch impact				
Type	As-welded		Stress-relieved	
	Energy min. avg. (ft-lbs)	Temp. (°F)	Energy min. avg. (ft-lbs)	Temp. (°F)
MIL-7018-M [6/]	50	-20	35	-20
MIL-8018-C3 [7/]	20	-20	20	-20
MIL-8018-C3SR [7/]	----	----	20	-20
MIL-9018-M [7/]	20	-60	20	-60
MIL-11018-M [7/]	35	-60	20	-60

NOTES:

1/ These are the minimum and maximum averages determined by testing all specimens required from each qualification or conformance test weld.

2/ Yield strength shall be determined by offset method (offset = 0.2 percent).

3/ Percent reduction in area shall be reported for information only.

4/ The upper value of the yield strength may be 5,000 psi higher than the indicated value for the $^{3}/_{32}$-inch size electrode.

5/ The ultimate tensile strength shall be recorded for information only.

6/ At the test temperature as specified in table III, ±3 °F, the average value of 5 tests shall be equal to or greater than the minimum average value specified. No more than two specimens may have values below the minimum average specified, but each specimen must have a value greater than or equal to 20 ft-lbs.

7/ Impact properties for the five specimens for each test condition shall be obtained at test temperatures as specified in table III herein, ±3 °F. The lowest impact value thus obtained together with the highest value shall be disregarded for this test. Two of the three remaining values shall be equal to or greater than that specified herein. One of the three remaining values may be up to 5 ft-lbs less than that specified herein. The computed average value of the three remaining values shall be equal to or greater than that specified herein.

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3.5 Chemical composition of deposited weld metal. Chemical composition testing of deposited weld metal shall be performed as specified in MIL-E-22200, with acceptance criteria per table IV (see 6.2).

TABLE IV. Chemical composition (percent) of deposited weld metal. 1/

Element	MIL-7018-M	MIL-8018-C3 MIL-8018-C3SR	MIL-9018-M	MIL-11018-M
Carbon	0.10	0.12	0.10	0.07 2/
Manganese	0.40 to 1.60	0.40 to 1.25	0.60 to 1.25	1.30 to 1.80
Silicon	0.80	0.80	0.80	0.60
Phosphorus	0.030	0.030	0.030	0.030
Sulfur	0.020	0.030	0.030	0.030
Chromium	0.15	0.15	0.15	0.40
Nickel	0.25	0.80 to 1.10	1.40 to 1.80	1.25 to 2.50
Molybdenum	0.35	0.35	0.35	0.25 to 0.50
Vanadium	0.05	0.05	0.05	0.05
Copper	3/	3/	3/	3/
Boron	4/	-----	-----	4/

NOTES:
1/ Single values are maximum percentages.
2/ On sizes of MIL-11018-M electrodes over $^{3}/_{16}$ in diameter the maximum carbon content is 0.10 percent.
3/ Use of the suffix RC, e.g., MIL-11018-M RC, indicates that the maximum copper content of the deposited weld metal shall be 0.15 percent. The remaining basic compositional requirements remain unchanged for the specific MIL-type. Report copper content for information only for electrodes without the RC suffix.
4/ Report for information.

3.6 Electrode identification. Electrodes shall be identified as specified in MIL-E-22200.

3.6.1 Heat and lot identification. When specified (see 6.2), each electrode shall be marked with heat or lot identification as specified in MIL-E-22200.

3.7 Electrode lengths. The lengths of electrodes shall be as specified in table V, except for Type MIL-8018-C3SR, which may be qualification tested in longer lengths (see 6.2).

TABLE V. Electrode lengths.

Sizes (inches)	Electrode lengths end-grip (inches)
$^{3}/_{32}$ 1/	9, 12, or 14
$^{1}/_{8}$, $^{5}/_{32}$, $^{3}/_{16}$	14
$^{7}/_{32}$, $^{1}/_{4}$, $^{5}/_{16}$	18

NOTES:
1/ See 6.2

3.8 Core wire alloy identity. The core wire alloy identity requirements shall be as specified in MIL-E-22200.

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3.9 Nondestructive inspection acceptance criteria. All nondestructive inspection of weld test assemblies (VT, RT, and MT) shall be performed to the requirements of T9074-AS-GIB-010/271 using acceptance criteria in MIL-STD-2035. All test weld assemblies shall meet Class 1 requirements of MIL-STD-2035 for VT and RT and the weld acceptance criteria for MT.

3.10 Instrumentation. Voltage shall be measured between the electrode holder and the test plate. Welding current and arc voltage measurements shall be made using analog or digital meters. RMS meters shall be used where any AC welding is performed. Tong meters shall not be used unless they are RMS-type and used for AC welding. Instrument calibration records shall be traceable to calibration performed as specified in ANSI/NCSL Z540.1 or equivalent standard.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be performed on iron-powder, low-hydrogen types of mineral covered electrodes when qualification is required (see 3.1). This inspection shall include the examination specified in 4.4.

4.3 Conformance inspection. Conformance inspection shall include the examinations specified in 4.5.3.

4.4 Qualification tests. Qualification tests shall be as specified in MIL-E-22200 and tables VI, VII, and VIII herein. Types MIL-7018-M, MIL-8018-C3, MIL-9018-M, and MIL-11018-M shall be tested in both the as-welded and stress-relieved conditions. Type MIL-8018-C3SR shall be tested in each of the stress-relieved conditions as specified in 3.4.1.a and b. MIL-8018-C3SR shall be tested with the longest length electrode intended to be used (see 3.7).

TABLE VI. Summary of required qualification and conformance inspection weld tests. [1/]

Weld joint design procedure	MIL-type	Welding position [2/]	Diameter sizes (inches)						
			3/32	1/8	5/32	3/16	7/32	1/4	5/16
Figure 1 and notes including table XII	All [3/, 4/]	F	X	X	X	X	X	X	X [5/]
Figure 2 and notes including table XIV	All [6/]	F	----	----	----	X	X	X	X [5/]
		V	X	X	X	----	----	----	----

NOTES:

1/ For qualification testing of the electrodes, all electrode types shall be tested using direct current electrode positive (DCEP). In addition, MIL-8018-C3 electrode shall be tested using alternating current (AC). For conformance inspection, all electrode types shall be tested using DCEP.

2/ V = vertical; F = flat.

3/ For MIL-11018-M electrodes, notes 2 and 4 to figure 1 do not apply and table XII does not apply. All other notes to figure 1 apply to MIL-11018-M. Unless otherwise specified (see 6.2), refer to table XIII for MIL-11018-M weld parameters and groove weld preparation requirements.

4/ For MIL-7018-M electrodes, the mechanical property test weldment shall be fabricated using the flat position with 3/16 inch and larger diameter electrodes and the vertical-up position with 5/32 inch and smaller diameter electrodes.

5/ Not applicable to MIL-9018-M and MIL-11018-M.

6/ Not applicable to MIL-7018-M.

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TABLE VII. Summary of required nondestructive (NDT) and mechanical tests for qualification and conformance inspection.

Figure No. [1/]	NDT [2/]			Tensile [3/]	Charpy	Chemical analysis	Requirements
	VT	MT [4/]	RT				
1	X	X	X [5/]	X	X	X [6/]	MIL-E-22200 and tables III and IV herein
2	X	----	X [5/]	----	----	----	

NOTES:

1/ Figures 1 and 2 apply to all electrodes.

2/ See 3.9.

3/ Includes ultimate, yield, elongation, and reduction in area.

4/ MT is not required for figure 1 test plates prepared using MIL-7018-M, MIL-8018-C3, MIL-8018-C3SR, and MIL-9018-M electrode.

5/ When both figures 1 and 2 are applicable, radiography of the figure 1 test assembly will satisfy the radiography requirements for figure 2 assemblies for Class 2 and 3 electrodes. If the figure 1 assembly is radiographed, the figure 2 test assembly may be eliminated for Class 2 and 3 electrodes.

6/ If chemical analysis is not determined using chemical analysis test pads welded as specified in MIL-E-22200, chemical analysis shall be determined from milling sample taken from the centerline of the broken tensile specimens or drillings, or by optical emission spectroscopy on any sample that is taken and analyzed at the depth of the tensile specimen centerline. In event of variances with composition requirements or conflicting results by testing activities, analysis shall be verified by procedures as specified in MIL-E-22200.

TABLE VIII. Summary of other tests.

Tests	Procedures	Requirements
Visual and dimensional	MIL-E-22200	MIL-E-22200 and 3.7 herein
Concentricity	MIL-E-22200	MIL-E-22200
Dielectric strength [1/]	MIL-E-22200	MIL-E-22200
Flaking and cracking of covering [1/]	MIL-E-22200	MIL-E-22200
Covering moisture	MIL-E-22200 and 4.8 herein or AWS A4.4M and 4.8 herein	3.3.1 herein
Total iron content in covering [1/]	MIL-E-22200	3.3.2 herein
Alloy identity	MIL-E-22200	MIL-E-22200
Diffusible hydrogen [2/]	4.6.1 herein	3.3.1 herein

NOTES:

1/ Required for qualification inspection only.

2/ Required for MIL-7018-M and MIL-11018-M type only.

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4.5 Conformance inspection.

4.5.1 Lot.

4.5.1.1 Inspection Level C. A lot of electrodes shall be as specified in MIL-E-22200 for inspection Level C for Types MIL-8018-C3 and MIL-8018-C3SR.

4.5.1.2 Inspection Level B. A lot of electrodes shall be as specified in MIL-E-22200 for inspection Level B for Types MIL-7018-M, MIL-9018-M, and MIL-11018-M.

4.5.2 Sampling for conformance inspection. Sampling for conformance inspection shall be as specified in MIL-E-22200 for those tests required in MIL-E-22200 and as specified in 4.5.3 for conformance weld tests.

4.5.3 Conformance inspection. Conformance tests shall be conducted on samples selected as specified in 4.5.2. One sample representing each lot shall be subjected to all the tests required by table IX. Samples representing each of the remaining wet mixes of covering mixture in a lot shall be tested as specified in 4.5.3.1.

TABLE IX. Conformance inspection.

Types	Tests	Procedures and requirements
All	Weld tests 1/, 2/	Tables VI and VII
	Other tests	Table VII
	Chemical analysis	Tables IV and VII

NOTES:

1/ Type MIL-8018-C3SR shall be tested in the stress-relieved condition only using treatment as specified in 3.4.1.a or b (see 6.2).

2/ When specified (see 6.2), Types MIL-7018-M, MIL-9018-M, and MIL-11018-M shall have the SR suffix added to the designation, and conformance inspected in the stress-relieved condition only, as specified in 3.4.1.a or c.

4.5.3.1 Wet mix equivalency tests. Weld deposits shall be analyzed as specified in table X, following the procedures of MIL-E-22200, with acceptance criteria of table IV herein. Wet mix equivalency tests are not needed when the lot is identified by controlled covering mixture chemical composition as defined in MIL-E-22200. The alternate chemical analysis method of MIL-E-22200 may be used for these tests.

TABLE X. Wet mix equivalency tests.

Types	Tests
MIL-8018-C3, and MIL-8018-C3SR	1/
MIL-7018-M, MIL-9018-M, and MIL-11018-M	2/

NOTES:

1/ Additional wet mix testing may be required as specified in MIL-E-22200.

2/ Required chemical analysis of the deposited weld metal is for selected elements only as specified in table XI.

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TABLE XI. Wet mix equivalency weld deposit chemical analysis.

Element	MIL-7018-M	MIL-9018-M	MIL-11018-M
Manganese	X	X	X
Nickel	----	X	X
Molybdenum	----	X	X
Copper	1/	----	----

NOTE:
1/ Analysis for copper is only required for electrodes with "RC" suffix.

4.5.3.2 Unsatisfactory test results. Retest shall be in accordance with the following.

4.5.3.2.1 Retests. When any original required conformance test (chemical, mechanical property, nondestructive inspection, visual/dimensional, etc.) representing a lot of material fails to meet specification requirements, the lot may be retested at the manufacturer's discretion. If the manufacturer chooses to retest the lot, the retests shall be as follows:

a. If the test failed due to one (or more) of the conditions listed in 4.5.3.2.1.1, then the lot shall be retested according to the requirements in 4.5.3.2.1.2.

b. If the test failed due to conditions other than those listed in 4.5.3.2.1.1, then the lot shall be retested according to the requirements in 4.5.3.2.1.3.

Note that all the specimens in a single test, whether the test involves a single specimen (such as for RT) or multiple specimens (such as for CVN impact tests), shall be considered a "set".

4.5.3.2.1.1 Testing/specimen preparation error retests. Retests to correct errors in testing and/or test specimen preparation are permitted under the following conditions:

a. Improperly fabricated test welds. This applies to test welds that have not been fabricated in accordance with the applicable specification or the weld procedure for the product under test.

b. Improperly prepared specimens. This applies to test specimens that are not in compliance with applicable standards.

c. Test equipment malfunction in the process of conducting a test.

d. Flaws in test specimens that are not indicative of inferior or defective product. Flaws at weld starts and stops at the ends of the test assembly, lack of fusion at the root and slag due to operator technique are normal weld artifacts and do not count as flaws indicative of inferior or defective products. Cracks, clustered porosity, flakes, and ruptures are flaws that may be indicative of inferior or defective products and shall not qualify a lot for a testing/specimen preparation error retest.

4.5.3.2.1.2 Testing/specimen preparation error retest requirements. Retests for conditions listed in 4.5.3.2.1.1 shall be as follows:

a. For each original test that failed due to one (or more) of the conditions in 4.5.3.2.1.1, only one testing/preparation error retest is permitted.

b. A testing/preparation error retest shall consist of one replacement specimen for each specimen that failed due to the conditions in 4.5.3.2.1.1. Note for VT, MT, and RT, the specimen is the entire weld. If the testing error is due to a malfunction of MT and/or RT equipment or to a flaw not indicative of inferior or defective products (see 4.5.3.2.1.1.d) removed by grinding or other suitable methods within $^{1}/_{16}$ inch of the surface, the original test assembly may be retested.

c. Results of all retests and reasons for all retests shall be reported on the certification of conformance.

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d. Retests of weld metal (e.g., mechanical properties) shall be from the same weld used for the original test, if feasible. If replacement test welds are needed, they shall be fabricated using consumables from the original sampling, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot and/or production run.

e. Retests of product (e.g., diffusible hydrogen, moisture, and chem. pads) shall be from the same sampling as the original test, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot and/or production run.

f. If the results of the testing/preparation error retest do not satisfy the applicable requirements, then a conformance retest shall be performed per 4.5.3.2.1.3.

4.5.3.2.1.3 Conformance retest requirements. Retests for failure to meet test requirements shall be as follows:

a. For each original test that failed due to conditions other than those listed in 4.5.3.2.1.1, only one conformance retest is permitted.

b. A conformance retest shall consist of repeating the test that failed two times (i.e., two-for-one). Note that a mechanical property test such as tensile or impact toughness involves a set of test specimens and therefore two full sets of specimens for each test that failed are required under this two-for-one rule. Results of all retests and reasons for all retests shall be reported on the certification of conformance.

c. Retests of weld metal shall be from the same weld used for the original test, if feasible. If replacement test welds are needed, they shall be fabricated using consumables from the original sampling, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot and/or production run.

d. Retests of product shall be from the same sampling as the original test, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot and/or production run.

e. Unless the welding procedure was determined to be the root cause of the original test failure, any replacement test weld should employ the same welding procedure as the original test weld. If a new procedure is used to correct the root cause of the original test failure, the retest shall include all tests conducted for the original weld. Two-for-one is only required for the test that failed. All future lots of the material shall use the new procedure. The reason for changing procedure shall be justified to the customer and included on the certification.

f. If the results of any conformance retest do not satisfy the applicable requirements, then the lot shall be rejected.

4.5.3.3 Check mechanical testing tolerances. Unless otherwise specified (see 6.2), retest as part of receipt inspection by a shipyard or other user of a product supplied in conformance with this specification will use the following mechanical property tolerances to verify acceptability of the product. If these mechanical property tolerances will not be used to verify acceptability of the product in retest as part of receipt inspection, this shall be specified in the order as indicated in section 6.2.

a. Yield strength test results (round off to nearest 1000 psi) may be 3 percent higher than the specified maximum or 3 percent lower than the specified minimum requirements of table III.

b. Ultimate tensile strength test results (when required) may be lower than the specified requirement of table III by 3 percent (rounded off to nearest 1000 psi).

c. Elongation test results shall not be lower than 93 percent of the minimum requirement specified in table III.

d. Charpy V-notch test results (both the average and the individual minimum value) may be lower than the requirements specified in table III by 10 percent.

e. No tolerance for the maximum yield strength requirement is allowed for the $^{3}/_{32}$-inch diameter MIL-7018-M electrode.

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4.5.4 Certification of conformance. Unless otherwise specified (see 6.2), a certificate of conformance shall be prepared for each lot of material offered for acceptance in accordance with the lot definition as specified in 4.5.1. The certificate shall include all actual data of all the chemical and mechanical tests as required by MIL-E-22200 and herein, including all test results that failed to meet those specified requirements, and a record of any stress relief treatment (if applicable). Operational parameters for any welded test assemblies shall also be reported. The manufacturer, customer, MIL-type, heat or lot, size and form shall be identified. Qualitative results of nondestructive tests, wet-mix equivalency test (if applicable), and other inspections or tests shall be recorded on the certificate. The certificate shall state that each lot has been sampled, tested, and inspected in accordance with the specification requirements to produce an electrode that meets the requirements within the tolerances allowed throughout the product. The certificate shall state that each lot meets all specification requirements, and shall be signed by a responsible representative of the manufacturer. Where test certificates issued by the manufacturer contain the above data requirements, a separate certificate of conformance will not be required. A certificate of conformance or equivalent test certificate shall be furnished to the consignee with the shipment.

4.6 Test procedures.

4.6.1 Diffusible hydrogen. For MIL-7018-M and MIL-11018-M electrodes, a sample of every lot of each size and MIL-type of electrodes furnished to the requirements of this specification shall be tested for diffusible hydrogen level in the weld metal. The test results shall be recorded along with the corresponding covering moisture. The hydrogen level of the deposited weld metal shall be determined by the analytical method of gas chromatograph or by the method of collection over mercury as specified in AWS A4.3. The test data shall be reported as specified in AWS A4.3.

4.6.2 Weld tests. Tables VI and VII summarize the weld tests required under this specification for qualification testing of electrodes and for the conformance inspection as specified in table IX. Ordinary-strength hull structural steel or higher-strength hull structural steel base material as specified in MIL-S-22698 and ASTM A36/A36M plate may be used for all electrode tests, except for MIL-11018-M electrodes. For MIL-11018-M type electrodes, the base plate material shall be HY-80 steel as specified in NAVSEA Technical Publication T9074-BD-GIB-010/0300. The plate shall meet the chemical composition requirements of NAVSEA Technical Publication T9074-BD-GIB-010/0300, and can be in the as-rolled condition without being quenched and tempered or tested for mechanical properties. For MIL-7018-M, when specified (see 6.2), the base plate material shall be HSLA-80 or HSLA-100 steel as specified in NAVSEA Technical Publication T9074-BD-GIB-010/0300, and the preheat and interpass temperature shall be 60 °F minimum.

4.6.2.1 MIL-11018-M weld tests. For Type MIL-11018-M electrodes, the weld tests shall be as specified in table VI, footnote 3 and table XIII.

4.6.2.2 Calculation of welding heat input. Heat input (J/in) shall be calculated using the following formula:

$$\text{Heat Input (J/in)} = \frac{\text{Volts x Amperes x 60}}{\text{Travel Speed (inches per minute)}}$$

The root pass(es) shall be excluded from all heat input calculations. Specified heat inputs reflect minimum acceptable ranges and qualification or quality conformance testing at values beyond the specified range shall be accepted.

4.6.2.2.1 Operational heat input calculation. Operational heat input shall be the actual heat input recorded or calculated for individual passes during welding of the test assemblies. For shielded metal arc welding, calculate the operational heat input of each pass in the test assembly using the following procedure:

a. To determine the travel speed for each complete weld pass, divide the total length of the complete weld pass by the total arc time for the pass. Total length of the complete weld pass and total arc time for the pass shall be calculated as in b and c below, respectively.

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b. On MIL-11018-M to determine the total length of the complete weld pass, measure each weld bead individually and add the bead lengths. Measure each bead from the apparent trailing edge of the beginning of the bead to the apparent trailing edge of the stop crater. On other electrode types the total completed weld pass may be measured to determine length, or alternately, the method described for MIL-11018-M may be used. Do not use the length of the test assembly as the pass length.

c. To determine the total arc time of the complete weld pass, measure the arc time for each weld bead in the pass and add the individual bead arc times together. An electronic arc timer or equivalent automated system that measures the total arc time for the pass may also be used.

d. Measure the arc voltage and welding amperage for each weld bead in the pass. Average the voltage and amperage to derive the average voltage and amperage values for the weld pass. If no adjustments are made to the output of the welding equipment, it is not necessary to measure the amps and volts on each bead.

e. Insert the average travel speed, arc voltage, and amperage for each complete pass into the heat input equation of 4.6.2.2 to calculate the heat input for the weld pass.

4.6.2.2.2 Average heat input. The average heat input shall be the average of the operational heat input of all passes in the test assembly and shall fall within the range (and tolerances) specified in the notes to figure 1 and table XIII.

4.6.3 Other tests. Table VIII summarizes tests, in addition to those as specified in table VI and table VII, required for qualification testing of the electrodes for the QPL and for conformance inspection, as applicable.

4.7 Core wire alloy identity tests. The core wire alloy identity test requirements shall be as specified in MIL-E-22200.

4.8 Temperature and humidity environmental exposure procedures. The electrode sample shall be exposed to elevated temperature and humidity conditions in accordance with exposure procedures specified in AWS A5.5/A5.5M. At the end of the exposure time, the electrodes shall be removed from the chamber and a sample of the coating taken for moisture determination as specified in the section entitled “Sample Preparation, Handling, and Storage” section of AWS A4.4M for covered electrodes.

4.8.1 Frequency of testing. The test of 4.8 shall be conducted for each lot, except that after testing ten consecutive lots of a specific product and size electrode with satisfactory results, the frequency of testing for that specific product and size electrode may be reduced to one lot in every ten lots. However, one unsatisfactory test result shall require reversion to the above cycle.

4.9 Chemical analysis. Welding of chemical analysis test pads shall be as specified in MIL-E-22200. The chemical compositions as specified in table IV are minimum and maximum limits. However, this specification does not imply and it should not be inferred that each of the possible chemistry compositions defined by these limits will meet the full requirements of this specification, for example, mechanical properties. Once a formulation is developed and qualified, and the aim values (composition validated by qualification testing) for each major alloying element have been established, changes to the formulation or aim values shall be verified not to alter the design criteria and be supported by test data. This information shall be reported to NAVSEA, who will determine if requalification is required in cases where the changes are found to adversely affect design criteria or performance.

4.10 Mechanical tests. Mechanical tests of welded test assemblies shall be as specified herein.

4.10.1 Impact test requirements.

4.10.1.1 Charpy V-notch testing. The Charpy V-notch specimens shall be machined to dimensions as shown in figure 1 and tested as specified in AWS B4.0. The notch shall be perpendicular to the plate surface. Impact properties for the five specimens shall be obtained at each test temperature as specified in table III ±3 °F.

4.10.2 Tensile testing. Two tensile test specimens shall be prepared and tested per AWS B4.0.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in MIL-E-22200.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification is intended to cover several low-hydrogen types of arc-welding electrodes of various alloys and strength levels, capable of depositing weld metal that, in the as-welded or stress-relieved conditions, meets mechanical property requirements given herein. It will be noted that, although these electrodes may be used for some applications in the as-welded condition, they are capable of stress relief as required. Properties may be affected by long time stress-relieving and slow cooling from stress-relieving temperature. Properties should be determined to be satisfactory after a simulated production postweld heat treatment by the user when required by the fabrication document. The designation, low hydrogen, is applied to electrodes having coverings that are substantially free of any compound that may liberate hydrogen in the arc. Low-alloy steel electrodes that produce low hydrogen in the arc have been found suitable for welding many steels that formerly gave serious difficulty through excessive porosity and underbead cracking, unless welded with high-alloy electrodes.

6.1.1 Type MIL-7018-M. This type is intended to cover medium steel electrodes used in welding medium or high-tensile steels and for attachment of such steels to high yield strength steels such as HY-80/100 and HSLA-80/100. This electrode is intended for use with DCEP electrical characteristics only.

6.1.2 Type MIL-8018-C3. This type is intended to cover low-alloy electrodes capable of depositing weld metal of higher strength than Type MIL-7018-M.

6.1.3 Type MIL-8018-C3SR. This type is intended to cover a special MIL-type electrode formulated to meet the long time (50 hours) stress-relieved mechanical properties requirements only.

6.1.4 Type MIL-9018-M. This type is intended to cover low-alloy electrodes suitable for welding steels having properties similar to those of the weld, and for welding attachments to HY-80 steel, as may be permitted. This electrode is intended to be used with the DCEP electrical characteristics only.

6.1.5 Type MIL-11018-M. This type is intended to cover low-alloy electrodes for welding groove butt joints in HY-80, HSLA-80, HSLA-100, and HY-100 steels where approved. This electrode is intended to be used with DCEP electrical characteristics only.

6.1.6 All types with suffix RC. This type is intended to cover electrodes for welding pressure and strength welds which will be subject to irradiation.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Whether stress-relieving treatments are required (see 3.4.1).

d. Whether the copper content is restricted as designated by the suffix RC to the MIL-type (see 3.5).

e. Whether electrode identification includes heat or lot identification (see 3.6.1).

f. For MIL-8018-C3SR, the length required (see 3.7).

g. For $^{3}/_{32}$-inch size electrode, the length required (see table V).

h. For MIL-11018-M, when parameters other than those of table XIII are to be used (see table VI, footnote 3, and table XIII, footnote 3).

i. For MIL-8018-C3SR, the stress-relieved condition required for conformance inspection (see table IX, footnote 1).

j. Whether Types MIL-7018-M, MIL-9018-M, or MIL-11018-M are to be conformance inspected in the stress-relieved condition (see table IX, footnote 2).

k. Whether mechanical property tolerances will not be used in retest as part of receipt inspection (see 4.5.3.3).

l. Whether HSLA-80 or HSLA-100 steel and a 60 °F preheat/interpass temperature is used for evaluating MIL-7018-M electrodes (see 4.6.2, and figure 1, footnote 3).

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m. If alternate restraint methods than fixturing are acceptable to minimize angular distortion of test welds (see table XIII, footnote 2).

n. If welding is to be performed using other than DCEP (see figure 2, footnote 1).

o. If certification of conformance is not required (see 4.5.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. QPL-22200-65 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil.

6.4 Subject term (key word) listing.

Heat input

Hydrogen, diffusible

MIL-7018-M

MIL-8018-C3

MIL-8018-C3SR

MIL-9018-M

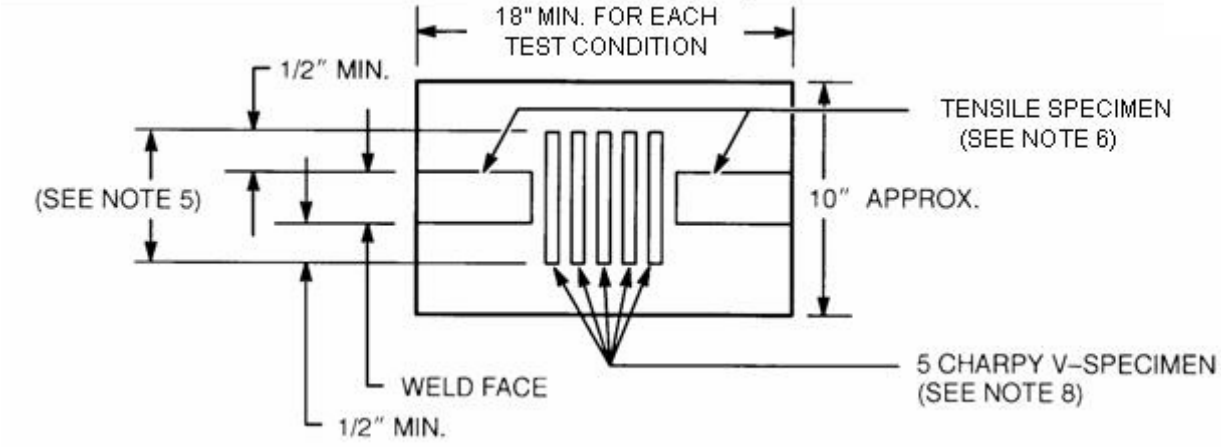
MIL-11018-M

Moisture, coating

Wire, core

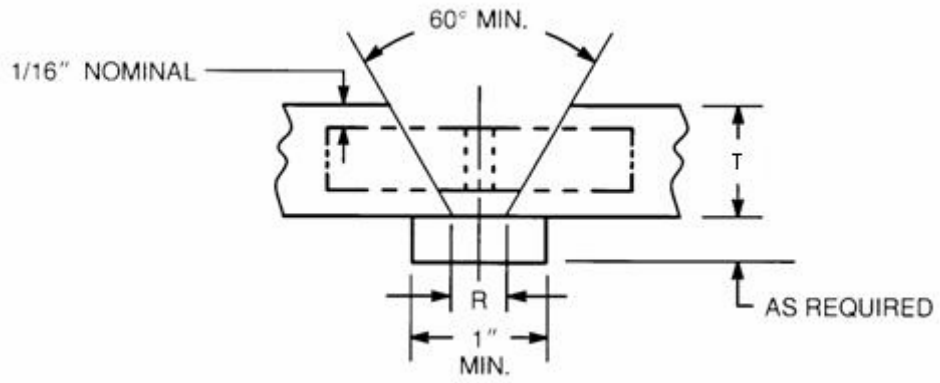
6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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LAYOUT OF TEST PLATE SHOWING LOCATION OF TEST SPECIMENS

NOTE: Each test condition refers to as-welded or stress-relieved conditions.



GROOVE PREPARATION OF TEST PLATE
SHOWING ORIENTATION OF CHARPY-V TEST SPECIMEN

FIGURE 1. Welded joint for mechanical and impact tests (test no. 3).

NOTES:

1. Base metal shall be as specified in 4.6.2.
2. Welds shall be deposited in layers using a 2½-core wire diameter maximum weave build-up sequence. Record and report as specified in table XII. Layers should be approximately ⅛-inch thick, and shall be deposited in such a manner that the final layers include a reinforcement of standard proportions ($^{1}/_{32}$ inch to ⅛ inch). Welding currents shall be as specified in the manufacturer's weld procedure and the actual values used shall be recorded and reported. Each layer shall be started at the finishing end of the preceding layer. This note does not apply to MIL-11018-M and MIL-7018-M electrodes.
3. For MIL-7018-M electrodes, the groove preparation of the test plate shall be a 45-degree included angle; the backing strap shall be ½ inch nominal thickness by 1 inch minimum width; the test plate width shall be 15-inch minimum and the length shall be 18-inch minimum; the heat input shall be 50 to 60 kilojoules per inch (kJ/in) for the ⅛-inch and larger diameter electrodes and 30 to 40 kJ/in for the $^{3}/_{32}$-inch diameter electrode; and the split-weave technique (the practice of depositing only the two wide-weave beads per weld layer) shall be limited to the first ½T of the weld joint thickness, and each weld layer beyond the first ½T of weld joint

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thickness shall contain three or more weld beads. The maximum reinforcement for the final layer shall not exceed $^{3}/_{16}$ inch.

4. After the joint has been tacked, the assembly shall be preheated to 225±25 °F prior to welding. After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the range 225±25 °F. Any required cooling of test assemblies shall be done in still air at room temperature, and assemblies shall not be force cooled with compressed air or by any other means. If it is necessary to interrupt this procedure, and the assembly cools, the assembly shall be reheated to 225±25 °F prior to resuming welding. When specified (see 6.2), the minimum preheat/interpass temperature shall be 60 °F when using MIL-7018-M electrode for welding HSLA-80 or HSLA-100 steel. This note does not apply to MIL-11018-M electrodes.
5. After completion of the weld and VT inspection is conducted, the weld reinforcement and backing strips shall be removed flush with the base plate on both surfaces and the assembly shall be subjected to the additional required nondestructive inspections. Weld pick-up (i.e., welding performed prior to formal nondestructive inspection to correct minor visual surface irregularities, such as low areas in the weld reinforcement, weld bead roll and a start-stop area) and minor repair (e.g., shallow lack of fusion against backing bar) may be accomplished on the weld face (before or after flush grinding or machining) and on the backside after backing bar removal provided that the same electrode being tested is used in conjunction with the same welding conditions, welding position, and other controls used in the original welding of the test assembly.
6. The weld shall be cut as shown resulting in two tensile coupons (i.e., blanks) suitable for 0.500-inch diameter specimens and a center coupon of size sufficient for the required Charpy V-notch specimens. No base metal shall be removed within ½ inch of the edges of the face of the weld by flame cutting. Only sawing or machining shall be used for specimen finishing.
7. One tensile coupon, where required, shall be removed from the stress-relieved welded test assembly as specified in 3.4.1 herein. The centerline of the all weld metal tensile specimens (as welded and stress relieved) shall be ⅜ inch below the top surface (or ¼ inch below the surface for ½-inch thick welds) of the test plate.
8. Both the stress-relieved and as-welded coupons, when required, shall be machined into 0.500-inch tensile specimens. For details see AWS B4.0.
9. The center section of the welded joint shall be used for the preparation of Charpy V-notch impact specimens. The Charpy V-notch specimens shall be machined as specified in AWS B4.0 and to the details shown on figure 1 with the exception that on the weld joints welded with the MIL-7018-M electrode the charpy impact specimens shall be taken from a location nominally 1/8-inch below the surface. Prior to machining specimens, where required, the welded test assembly shall be stress-relieved as specified in 3.4.1 herein.
10. Impact properties for the five Charpy V-notch specimens for each test condition shall be obtained at test temperatures as specified in table III herein, ±3 °F. The computed average value shall be determined as specified in the notes to table III and shall be greater than or equal to that as specified in table III herein.

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TABLE XII. Groove weld preparation. [1/] [2/]

Electrode size (inches)	Minimum thickness (T) (inches)	Maximum root opening (R) [3/] (inches)
3/32	¾	¼
⅛	¾	¼
5/32	¾	¼
3/16	¾	¼
7/32	¾	½
¼	1	½
5/16	1¼	½

NOTES:

1/ Does not apply to MIL-11018-M electrodes.

2/ Pass and layer sequence shall be reported. A sketch is acceptable, as is use of written format for bead placement, such as 1 full, 2 split, 3 triple, and 5 quad layers.

3/ Root opening may be increased to allow proper bead placement and avoid slag entrapment.

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TABLE XIII. MIL-11018-M welding requirements. 1/, 2/

Electrode size (inches)	Minimum thickness (T) (inches)	Maximum root opening (R) (inches)	Welding parameters	
			Heat input (KJ/in) 3/, 4/	Preheat/interpass temperature (°F) 5/, 6/
3/32	¾	½	30-35	200-225
⅛	¾	½	35-41	200-225
5/32	¾	½	45-51	200-225
3/16	¾	½	51-57	200-225
7/32	1	½	51-57	200-225
¼	1	½	51-57	200-225

NOTES:

1/ Pass and layer sequence shall be reported. A sketch is acceptable, as is use of written format for bead placement, such as 1 full, 2 split, 3 triple, and 5 quad layers. The final weld layers shall include a reinforcement of standard proportions (1/32 inch to ⅛ inch).

2/ Welded test assemblies shall be insulated or separated from the workbench. To minimize the angular distortion of the completed test weld, test assemblies shall be fully restrained during heating, welding and cooling by appropriate fixturing, unless otherwise specified (see 6.2). Restraining fixtures shall not interfere with the placement of strip heaters or heating pads on the backside of the assembly. Contact points between the fixture and the test plates shall be minimized to minimize the heat sink effect of the fixturing. Base plates either side of the weld joint shall not be back set (reverse distorted) before welding to compensate for weld shrinkage, as an alternative to the required fixturing.

3/ Alternate NAVSEA approved heat inputs and test plate thickness may be specified (see 6.2).

4/ See 4.6.2.2 for heat input calculation procedures.

5/ Preheat shall be applied evenly along the length of the plate using electric resistance contact strip heaters or electric resistance ceramic pad heaters. If temperature indicating crayons are used, the preheat and interpass temperature shall be measured at several points along the entire length of weld 1 to 1½ inches from the edge of the bevel on both plates being joined. If a contact pyrometer is used, the preheat and interpass temperature shall be measured in at least 3 places 1 to 1½ inches from the bevel on each side of the plate (at each end and the middle). The heat shall be applied in a way to ensure that it is through thickness, with the heater strips or heater pads on the bottom of the plate (opposite from the side being welded).

6/ After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the preheat range as specified in table XIII. Any required cooling of test assemblies shall be done in still air at room temperature, and assemblies shall not be force cooled with compressed air or by any other means. If it is necessary to interrupt this procedure, and the assembly cools, the assembly shall be reheated to the preheat temperature, as specified in table XIII, prior to resuming welding.

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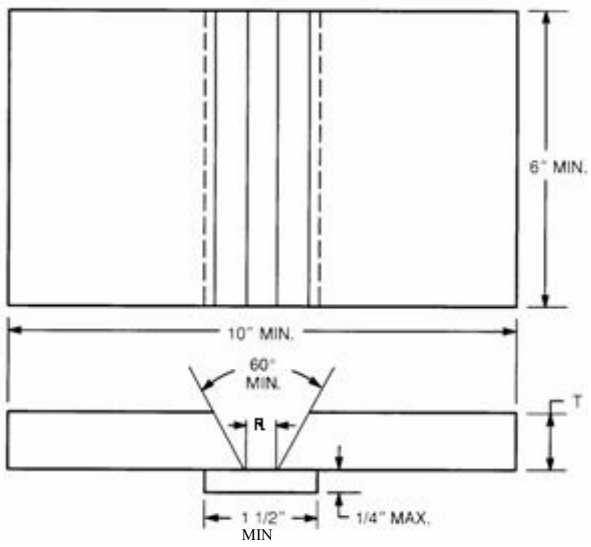


FIGURE 2. Groove weld usability test (test 8a).

NOTES:

1. Welding shall be accomplished with the test plate in the vertical or flat position as specified in table VI. Welding current shall be as specified in the manufacturer's weld procedure and the actual values shall be recorded and reported. Unless otherwise specified by the acquisition document (see 6.2), test shall be done with DCEP only.
2. Base metal shall be as specified in 4.6.2 herein.
3. The edge preparation if flame cut shall be ground to remove slag and surface irregularities.
4. Welds shall be deposited in approximately ⅛-inch thick layers. Weld bead widths shall be as shown in table XIV. Each pass shall contain a start in the area to be evaluated.
5. The as-welded reinforcement shall be $^{3}/_{16}$-inch maximum.
6. Preheat and interpass temperature shall be 225±25 °F.
7. After completion of the weld and conduct of VT inspection, the weld reinforcement and backing strips shall be removed flush with the base plate on both surfaces and the assembly shall be subjected to the additional required nondestructive inspections. Weld pick-up and minor repair may be accomplished on the weld face (before or after flush grinding) and on the backside after backing bar removal provided that the same electrode being tested is used in conjunction with the same welding conditions, welding position, and other controls used in the original welding of the test assembly.
8. Not applicable to MIL-7018-M.

TABLE XIV. Groove weld usability test preparation.

Electrode size	$^{3}/_{32}$ (inch)	⅛ (inch)	$^{5}/_{32}$ (inch)	$^{3}/_{16}$ - $^{5}/_{16}$ (inch)
Thickness (T) 1/	¼	⅜	½	½
Root opening (R) 2/	¼	$^{5}/_{16}$	⅜	½
Weld bead width (max)	⅜	½	⅝	¾

NOTES:

1/ Minimum.

2/ Tolerance ± $^{1}/_{16}$ inch.

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